

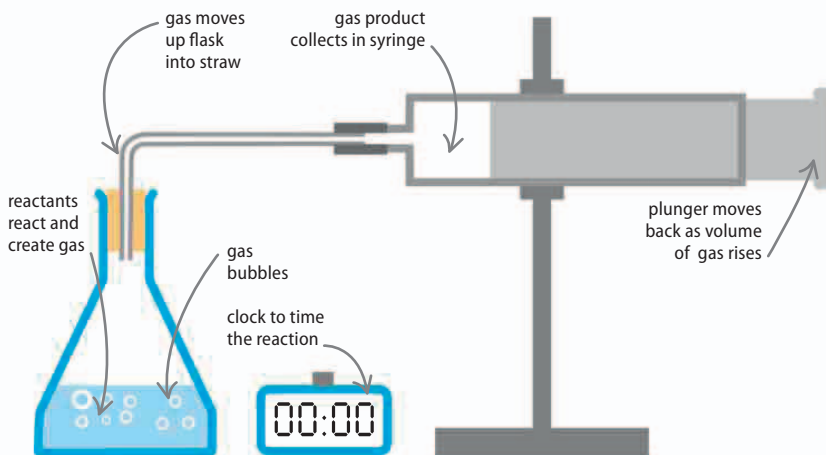
Rates of reaction

REACTANTS TURN INTO PRODUCTS AT DIFFERENT RATES.

Reaction rates depend on the substances involved. Dynamite burns so rapidly that it explodes in a fraction of a second, while an iron nail takes years to turn to rust.

Measuring rates

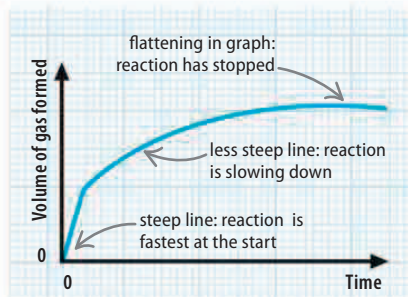
To understand what controls the speed of a reaction, a chemist first needs to be able to measure its rate—how quickly the reactants are converting into products. Only one product needs to be measured, since any others are produced at the same rate.



△ Using a syringe to measure gas

Measuring the increase in the volume of a gas product is relatively simple using a syringe. The measurements can be taken at regular intervals, timed by the clock. The volume will increase at a rate that is proportional to the reaction.

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△ Rate of reaction graph

The volume measurements can be plotted against time on a graph. The steep line at the beginning shows that the rate of reaction starts very fast but then tails off.

Reactivity and temperature

Every reaction has an activation energy, which is the amount of energy the reactants need in order to break and reform atomic bonds. When a reaction has low activation energy, more of the reactants have the amount of energy needed and so the reaction occurs more quickly than a reaction with a higher activation energy. Heating the reactants—and increasing the pressure—adds energy and increases the rate of reaction.

▷ Magnesium in water

In cold water, magnesium reacts very slowly, forming magnesium oxide and bubbles of hydrogen. Heating the water to near its boiling point makes the same reaction run more quickly, making the water fizz with hydrogen bubbles.

